

ThermalManhattan: Robust Thermal Compass under Radiometric Flatness in Manhattan World

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<https://ThermalManhattan.github.io/>

Abstract. We present a drift-free, matching-free 3-DoF visual compass for monocular thermal imagery. Thermal cameras sense emitted radiation and operate regardless of illumination, but when surfaces reach thermal equilibrium, their texture vanishes, so point features collapse and the genuine, shape-preserving cues survive only as structural lines at object boundaries. Existing line-based thermal approaches still match lines across frames, where lines are not extracted consistently from frame to frame and tracking further degrades as the lines become too sparse. We instead avoid inter-frame line matching and, to the best of our knowledge, are the first to bring the Manhattan world (MW) assumption to thermal imagery, estimating the absolute camera orientation independently for every frame. Given a gravity vector from the drone, we recover the remaining heading angle (yaw) from structural lines, so each frame is fixed on its own and no drift accumulates across NUC drops or aggressive motion. We collect six outdoor man-made UAV sequences and, since structure-from-motion is unreliable on monocular thermal, build pseudo ground-truth (GT) trajectories with RTK-GPS priors. We publicly release this benchmark, the first paired RGB–thermal night-time UAV dataset under the MW assumption. On this benchmark, the proposed method outperforms existing rotation estimators and stays stable over 20+ rotational loops on which the onboard gyro integrates 15° of yaw drift at the end.

Keywords: Rotation estimation · Manhattan World · Thermal imaging · UAV · Vanishing points · Benchmark

1 Introduction

Night-time UAV operation is increasingly central to surveillance, search and rescue, and infrastructure inspection. Conventional RGB pose estimation and SLAM are well studied but largely fail in low light, where little reflected light is available and few stable features can be extracted. Long-wave infrared (LWIR) thermal cameras instead sense emitted radiation and remain functional regardless of illumination, making them a suitable sensor for night-time UAV per-

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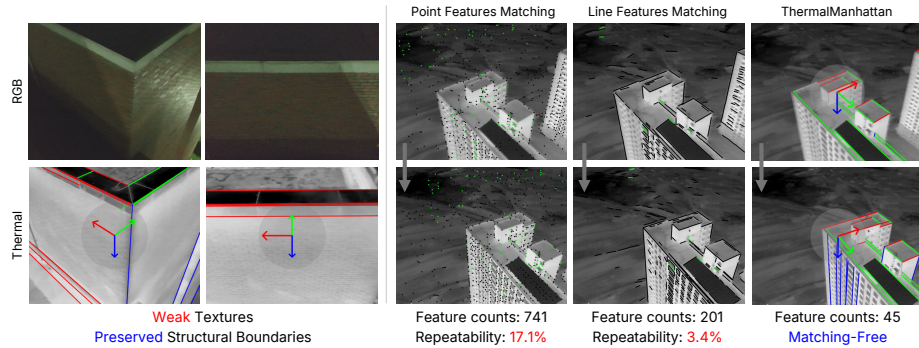


Fig. 1: ThermalManhattan under challenging night-time thermal conditions. **(Left)** RGB-thermal examples of radiometric flatness: RGB images are dark, while thermal images have weak surface texture but retain structural boundaries that support our rotation estimates. **(Right)** A challenging thermal frame pair with point and line repeatability of 17.1% and 3.4%, measured following [24] and [27], respectively. Green marks repeatable features; black marks non-repeatable ones. The corresponding averages over all six sequences are 64.2% and 41.5%. ThermalManhattan estimates the absolute 3-DoF orientation relative to the MW independently for each frame, without inter-frame feature matching.

ception [19]. Using thermal imagery for localization, however, is fundamentally harder than swapping the sensor, for a physical reason.

A thermal camera measures emitted radiation, so when a surface reaches *thermal equilibrium*, its temperature becomes uniform and its texture disappears almost entirely [4, 12, 23]. Relying on point features extracted from such surfaces for navigation is physically unreasonable and leads to inevitable collapse, an effect that persists even after on-sensor correction of fixed-pattern and low-frequency noise [33, 45]. The radiometric signal survives where it physically must: at occlusion edges and structural boundaries, where a real difference in emitted radiation appears across the discontinuity. In the thermal domain, the shape-preserving *genuine feature* is thus not the point but the *line*, which makes line-based cues the physically grounded choice for thermal navigation.

Recent thermal methods have recognized this and turned to edges and lines [12, 15, 18, 38]. However, line- and edge-based odometry and SLAM rely on inter-frame tracking or optical flow and inherit two thermal-specific failure modes. Non-uniformity correction (NUC) freezes the stream for 0.1–1.0s and produces abrupt viewpoint jumps that break inter-frame association [19], and thermal equilibrium can leave so few lines that the matches needed to propagate pose vanish. In either case, the sequential methods lose tracking, and the estimate easily collapses.

To address these issues, we exploit the Manhattan world (MW) model [5], under which the dominant structural lines of man-made scenes align with three orthogonal directions whose vanishing points localize the camera’s absolute orientation from a single frame, avoiding inter-frame line matching and tracking. Because each frame is solved on its own, a NUC jump only affects that frame

and never propagates, and orientation is still recovered when only one or two structural lines survive, exactly the extreme-sparsity regime where matching-based systems fail. To the best of our knowledge, this is the first drift-free and matching-free thermal visual compass with actual flight drone data. In summary, our main contributions are as follows:

- **The first drift-free and matching-free thermal compass.** Grounded in the physics of thermal equilibrium, we transfer the gravity-anchored Mine-and-Stab (MnS) yaw search of MWMS [14] to thermal imagery, yielding an absolute per-frame orientation that is robust to NUC drops, aggressive motion, and pure 360° rotation in place.
- **Structural line detector and refinement cost for thermal imaging.** We employ a wireframe parser, PLNet [39], which detects long structural lines despite noisy boundaries in the thermal imagery. A spherical residual replaces the image plane residual so that fragmented segments cannot be outvoted by long spurious ones, and a Cauchy loss suppresses outlier lines.
- **A paired RGB–thermal night-time UAV benchmark.** We evaluate ThermalManhattan on the author-collected airborne sequences captured by flying drones (low to extreme-low light, pure rotation to 6-DoF) with RTK-anchored pseudo ground-truth trajectories.

2 Related Work

2.1 Thermal Odometry and SLAM

Thermal imagery faces low contrast, fixed-pattern noise (FPN), automatic gain control (AGC) drift, and NUC frame drops. Prior work responds mostly by strengthening feature extraction and matching. TP-TIO [45] learns a thermal keypoint detector with IMU-aided tracking, TI-SLAM [11] bridges NUC drops with a lightweight learned flow, Self-TIO [19] self-supervises 16-bit feature extraction and tracking, and DeepTIO [32] and Graph-TI-SLAM [33] regress relative motion end-to-end. Since points are what radiometric flatness erases, a second family turns to boundary cues: ETIO [38] tracks binarized thermal edges with an adaptive KLT, PI-EAT-SLAM [12] couples edge tracking with thermal Gaussian-splatting mapping, and TPL-SLAM [15] tracks ELSEd lines with line optical flow. These methods focus on what to extract, such as points, edges, and lines, but do not change how to estimate: relative motion accumulated through frame-to-frame association, typically bridged by inertial propagation [31]. When NUC freezes the stream or boundaries become too sparse to match, the association breaks, and the estimate degenerates into dead reckoning or collapses. Our ThermalManhattan keeps the line-centric representation but discards inter-frame association by aligning structural lines with the three orthogonal MW directions, and it localizes the absolute camera orientation of every single frame. A NUC drop can corrupt at most one frame, no drift accumulates, orientation survives with only one or two lines, and the IMU serves solely as a gravity reference.

2.2 Visual Compass from Structural Regularities

Aligning the three orthogonal vanishing points (VPs) of man-made scenes with the camera frame has recovered an absolute rotation from a single image, with no inter-frame correspondence and no drift to accumulate. Subsequent work proposes this kind of visual compass on RGB imagery: branch-and-bound search [2], facility-location line clustering [1], quasi-globally optimal search [20], convex relaxation [22], and learned VP detection [47]. MWMS [14] anchors this search on a known gravity direction, reducing orientation to a real-time, quasi-optimal 1-DoF Mine-and-Stab (MnS) yaw search per frame, which our ThermalManhattan builds on. All of these methods, however, are designed for and evaluated on RGB imagery where structural lines are sharp and plentiful. Structural regularity rotation estimation has never been carried out on thermal imagery, the domain whose failure modes (NUC drops, low contrast, sparse boundaries) make a matching-free, drift-free rotation cue most valuable. We close this gap by combining RGB-pretrained wireframe parsing with a spherical cost, adapting rotation refinement to sparse, noisy thermal boundaries.

2.3 Thermal Odometry Datasets and Benchmarks

Table 1: Comparison of thermal odometry datasets. Ours is the only benchmark that simultaneously satisfies the UAV platform, MW environment with GT trajectory, and camera calibration.

Dataset	GT	Hardware	Environment	Manhattan World
TIO for flight [7]	✓	UAV	Outdoor	✗
MS [6]	✓	Handheld	Indoor, Outdoor	✗
OdomBeyondVision [21]	Δ^1	Handheld, UGV, UAV	Indoor	✓
MS ² [35]	✓	Driving	Outdoor	✗
ViViD++ [16]	✓	Handheld, Driving	Indoor, Outdoor	Δ^2
STheReO [42]	✓	Driving	Outdoor	✗
Caltech Aerial [17]	✓	UAV	Outdoor	✗
SubT-MRS [44]	✓	UGV	Indoor, Outdoor	✓
DiTer++ [13]	✓	UGV	Outdoor	✗
M2DGR [41]	✓	UGV	Indoor, Outdoor	Δ^2
TP-TIO [45]	✗	Handheld	Indoor	✓
TI-SLAM [11]	✓	Handheld	Indoor, Outdoor	Δ^2
FIReStereo [9]	✓	Handheld, UAV	Outdoor	✗
TPL-SLAM [15]	✗	Handheld	Indoor	✓
Ours	✓	UAV	Outdoor	✓

Δ^1 : provides GT trajectory, but doesn't release the thermal camera calibration file.

Δ^2 : captured in man-made environments, while other segments (object-facing views, open scenes) lack three recoverable orthogonal vanishing points.

Public thermal datasets span diverse platforms and scenes (Tab. 1), yet none of them support evaluating a UAV thermal visual compass in urban environments. Aerial datasets cover night-time flight [7], natural landscapes [17], and

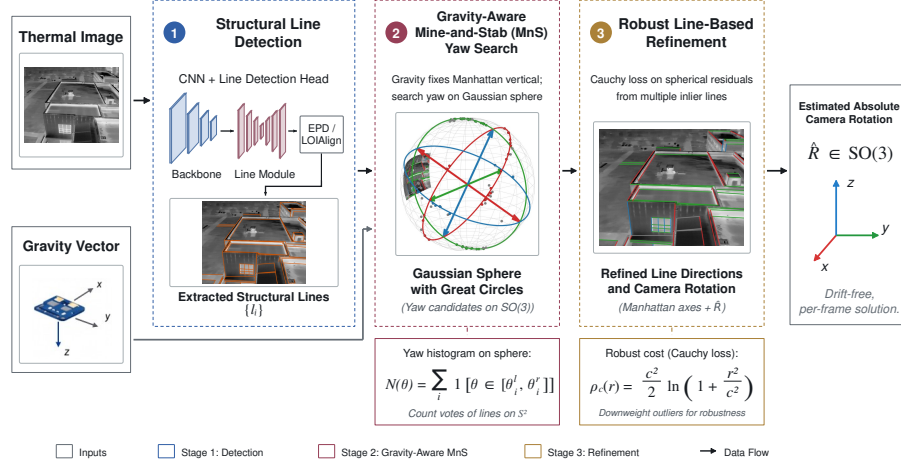


Fig. 2: Overview of ThermalManhattan. PLNet [39] extracts structural lines from each thermal frame. The onboard gravity vector anchors the Manhattan vertical, and the Mine-and-Stab (MnS) search from MWMS [14] resolves the single remaining DoF (yaw). Refinement with Cauchy loss on spherical residuals improves the yaw estimate. Every frame independently yields an absolute orientation without inter-frame line matching or tracking, so no drift error can accumulate.

smoke-filled forests [9] with little MW structure. [17] reports clipping its tracking sequences due to periodic feature sparsity. Outdoor driving and legged-robot datasets [13, 35, 42] cover both structured and unstructured environments, but are likewise non-MW. The datasets with man-made MW structure [11, 15, 16, 41, 44, 45] are handheld or ground-based, largely indoor, and several withhold true trajectories [15, 45]. OdomBeyondVision [21] includes UAV sequences but is restricted to indoor environments. No existing benchmark offers, at once, an aerial platform, an outdoor MW environment, released GT and calibration, and the aggressive rotational motion that stresses a thermal compass most. Our night-time RGB–thermal UAV benchmark with six outdoor flights from pure rotation to 6-DoF with RTK-anchored pseudo-GT fills this gap.

3 Methods and Datasets

ThermalManhattan adapts MWMS [14] to thermal imagery through per-frame line detection, gravity-anchored MnS yaw search, and robust spherical refinement (Fig. 2).

3.1 Preliminaries: Rotation from Lines in a Manhattan World

We first give the geometric intuition behind line-based rotation estimation, following the formulation of MWMS [14].

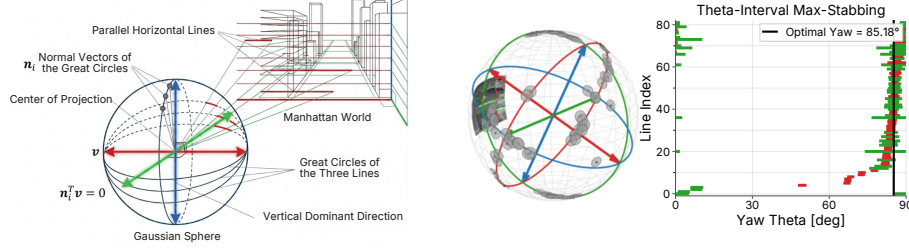


Fig. 3: Geometry of line-based MW rotation. **(Left)** Each image line maps to the normal of its great circle on the Gaussian sphere (gray dots); lines and their Manhattan direction share a color, and a line of direction $\mathbf{v}$ satisfies $\mathbf{n}_i^\top \mathbf{v} = 0$. Gravity fixes the vertical direction (blue). **(Right)** With the vertical fixed, each line defines a candidate yaw interval $[\theta_i^l, \theta_i^r]$. Red and green intervals correspond to the two mutually orthogonal horizontal MW directions. The 90° periodicity of the MW restricts the search to $[0^\circ, 90^\circ)$, and MnS returns the yaw $\hat{\theta}$ (black vertical line) that stabs the most intervals.

Vanishing directions and absolute rotation. Parallel 3D lines of one direction (e.g. vertical building edges) project to image lines that meet at a vanishing point, i.e. a 3D direction $\mathbf{v}$ expressed in the camera frame. A Manhattan world (MW) has three mutually orthogonal such directions, and stacking them as columns gives the rotation $\mathbf{R}_{cM} \in SO(3)$ from the MW frame to the camera frame. Because these directions are read from a *single* image, the estimated orientation is absolute, and no drift accumulates across frames.

Lines as great circles. It is convenient to reason on the Gaussian sphere, the unit sphere at the center of projection (Fig. 3, left). An image line together with the center of projection spans a plane that cuts the sphere in a great circle, compactly written by its unit normal $\mathbf{n}_i$. A line belongs to direction $\mathbf{v}$ exactly when its great circle passes through $\mathbf{v}$, equivalently

$$\mathbf{n}_i^\top \mathbf{v} = 0. \quad (1)$$

Estimating a direction thus amounts to finding the $\mathbf{v}$ orthogonal to the normals of the lines that belong to it.

Reducing to a single angle. The gravity vector from the attitude and heading reference system (AHRS) of drones gives the vertical Manhattan direction and fixes two rotational DoF (roll, pitch); only the yaw θ about the vertical remains. As θ varies, the two horizontal directions rotate around the fixed vertical, so the whole rotation is a function of this one angle, and rotation estimation becomes a 1-D search for the θ whose horizontal directions are most consistent with the line normals through Eq. (1).

3.2 Structural Line Detection

We employ PLNet [39], which extends a SuperPoint-style shared backbone [8] with a dedicated line detection branch. The branch predicts segments using the attraction-region representation of HAWPv2 [40], associates them with nearby

junctions to form wireframe proposals, and verifies them with a classifier using LOIAlign features. This suppresses fragmented detections and produces long, complete structural lines that preserve the dominant MW geometry. We use RGB-pretrained PLNet weights without fine-tuning on thermal imagery.

3.3 Rotation Estimation via Manhattan-World Mine-and-Stab

We determine the heading angle (yaw) with the Mine-and-Stab (MnS) search in MWMS [14] given roll and pitch angles from a gravity vector. For a line i assigned to a horizontal direction, the set of yaw angles under which it stays an inlier (its normal within a tolerance of orthogonality, $|\mathbf{n}_i^\top \mathbf{v}(\theta)| \leq \tau$) forms a contiguous *candidate interval* $[\theta_i^l, \theta_i^r]$ on the θ axis (Fig. 3, right). *Mining* collects these intervals from all lines; *stabbing* sweeps a probe along θ and counts how many intervals it pierces,

$$N(\theta) = \sum_i \mathbb{1}[\theta \in [\theta_i^l, \theta_i^r]], \quad \hat{\theta} = \arg \max_{\theta} N(\theta) \quad (2)$$

i.e. the yaw consistent with the largest number of lines. Since the two horizontal directions are 90° apart, the intervals repeat with a 90° phase shift and the search collapses to $\theta \in [0^\circ, 90^\circ)$, so a single sorted pass over the interval endpoints returns the global optimum in real time. This global, memoryless counting is what makes $\hat{\theta}$ insensitive to a handful of stray lines: a few outliers cannot outvote the dominant structural population.

3.4 Rotation Refinement with Multiple Lines

The MnS search returns an initial yaw estimate $\hat{\theta}$ that maximizes the inlier count. We further refine $\hat{\theta}$ by minimizing a robust cost over the inlier lines,

$$\hat{\mathbf{R}} = \arg \min_{\mathbf{R} \in \mathcal{M}(\theta)} \sum_{i \in \mathcal{I}} \rho(r_i(\mathbf{R})), \quad (3)$$

where $\mathcal{M}(\theta)$ is the gravity-fixed 1-DoF family of Manhattan rotations parameterized by yaw, $\rho(\cdot)$ is a robust loss, and r_i is the per-line residual between a line and its assigned VP direction $\mathbf{v}_i \in \{\mathbf{v}_1, \mathbf{v}_2\}$. Two choices define this stage: how r_i is measured and which loss ρ is used, and both differ from the original MWMS refinement [14].

Residual geometry. Each detected line l_i is assigned to the horizontal VP direction it is closest to being orthogonal with; a line is kept as an inlier only if this closeness is within a gate, $\min_k |\mathbf{n}_i^\top \mathbf{v}_k| < \sin 2^\circ$. Let $\mathbf{n}_i$ be the unit normal of the line’s great circle on the Gaussian sphere. MWMS measures the residual on the *image plane*, as the perpendicular pixel distance from the segment endpoints to the virtual line joining the segment midpoint and the projected VP $\pi(\mathbf{K}\mathbf{v}_i)$. For a segment of pixel length ℓ_i^{px} , let α_i denote the acute angle between the detected and virtual image lines. This distance scales as

$$r_i^{\text{img}} \propto \ell_i^{\text{px}} \sin \alpha_i. \quad (4)$$

We instead measure the residual directly on the sphere, as the deviation of the great-circle normal from orthogonality with the VP direction,

$$r_i^{\text{sph}} = \mathbf{n}_i^\top \mathbf{v}_i = \sin \varphi_i \in [-1, 1], \quad (5)$$

where φ_i is the signed angular deviation of $\mathbf{v}_i$ from the line’s great circle. Here, α_i is defined in the image plane, whereas φ_i measures the angular deviation on the Gaussian sphere. Both residuals vanish at perfect alignment, but (4) carries the factor ℓ_i^{px} while (5) does not; the consequences are analysed in Sec. 4.6.

Robust loss. With residual r and scale c , we compare the plain squared loss $\rho_{\text{L2}}(r) = \frac{1}{2}r^2$ against the redescending Cauchy (Lorentzian) M-estimator,

$$\rho_{\text{C}}(r) = \frac{c^2}{2} \ln \left(1 + \frac{r^2}{c^2} \right), \quad (6)$$

whose influence functions $\psi = \rho'$ are

$$\psi_{\text{L2}}(r) = r, \quad \psi_{\text{C}}(r) = \frac{r}{1 + r^2/c^2}. \quad (7)$$

The problem (3) is minimised by iteratively reweighted least squares with weights $w(r) = \psi(r)/r$. The weights depend only on the residual magnitude. With our spherical residual, no explicit line-length term enters either the residual or the IRLS weighting. For our final refinement, we apply the Cauchy loss to the spherical residual (5), with scale $c = \sin 1^\circ$, and use a single refinement pass. The ablation over residual geometry and loss, and the analysis of why this combination is preferred on thermal imagery, are given in Sec. 4.6.

3.5 RGB–Thermal Night-time UAV Dataset

As summarized in Sec. 2.3, existing thermal benchmarks lack the combination of aerial capture, outdoor MW scenes, reference trajectories, and camera calibration. We collected a paired RGB–thermal dataset using a DJI Mavic 3T, recording thermal images (640×512 , DFOV 61°) and RGB images (1920×1080 , DFOV 84°) at 30 Hz. It consists of six night-time sequences in man-made scenes that satisfy the MW assumption, spanning illumination from medium-dark to extreme-dark and motion from pure rotation to full 6-DoF, including short and long rotational trajectories and closed/open-loop 6-DoF flights as shown in Tab. 2.

Ground-truth protocol. Obtaining accurate rotation GT for outdoor night-time UAV thermal sequences is non-trivial: motion capture is infeasible over flight-scale baselines, onboard orientation estimates can drift, and the standard image-only alternative, COLMAP [34], is unreliable on thermal sequences [4, 23, 25, 46] because thermal correspondences are sparse and noisy, while NUC-induced frame freezes can further disrupt inter-frame association. These effects can fragment the reconstruction or produce duplicated structures. We therefore construct pseudo-GT using RTK-anchored COLMAP. RTK-GPS positions are used both to select spatially adjacent image pairs and as camera-center position priors during bundle adjustment. Fig. 5 compares image-only and RTK-anchored COLMAP, showing improved agreement with the RTK-GPS trajectory

Table 2: Composition of our self-collected dataset.

Sequence	Light Condition	Movement		Time
S-Buildings	Medium dark	Pure rotation	slow	40.4 s
N-Buildings	Medium dark	Pure rotation	fast, 5 loops	57.7 s
C-Buildings	Medium dark	Pure rotation	dynamic, 20+ loops	6 m 51 s
S1 Building	Medium dark	6-DoF motion	moderate-fast, circling	10 m 17 s
Residential-closed	Extreme dark	6-DoF motion	fast, loop closure	7 m 34 s
Residential-opened	Extreme dark	6-DoF motion	no loop	5 m 48 s

**Fig. 4:** Representative sequences of our benchmark. Each row shows RGB-thermal frame pairs for one sequence. The last column shows the camera motion: per-frame yaw for the pure-rotation N-Buildings sequence, and RTK-GPS trajectories color-coded by speed for S1 Building and Residential-closed.

and reduced structural duplication in the reconstruction. The resulting sparse reconstruction is used to estimate thermal-camera poses at 30 fps through PnP. RTK-GPS constrains camera positions, while orientations remain estimated from thermal-image correspondences; we therefore refer to these poses as pseudo-GT rather than externally measured rotation ground truth.

4 Experiments

4.1 Experimental Setup

We evaluate ThermalManhattan on the six sequences in our benchmark, using rotation-error metrics suited to each motion type.

Metrics. Following [14, 26], we evaluate Absolute Rotation Error (ARE) as

$$\text{ARE} = \arccos\left(\frac{\text{tr}(\mathbf{R}_{\text{gt}}^{\top} \hat{\mathbf{R}}) - 1}{2}\right), \quad (8)$$

where $\mathbf{R}_{\text{gt}}$ and $\hat{\mathbf{R}}$ denote the reference and estimated rotations, respectively. All errors are reported in degrees. We also evaluated ARE using the equivalent arcsin formulation [10], which is better conditioned near zero error, and obtained identical results at the reported precision.

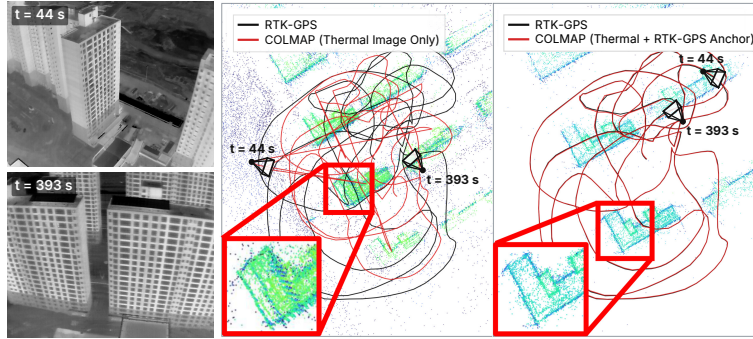


Fig. 5: (Left) Example thermal frames at $t = 44$ s and 393 s. (Middle) Image-only COLMAP reconstruction. (Right) RTK-anchored COLMAP reconstruction. Black and red curves show the RTK-GPS reference and estimated camera trajectories, respectively. RTK anchoring improves SfM reconstruction and supports more reliable 6-DoF pose estimation for pseudo-GT generation, as illustrated by the magnified regions.

Table 3: Closed-loop consistency error (CLE, in degrees) on the pure-rotation sequences S- and N-Buildings. The best result for each sequence is shown in bold.

Seq.	Matching-based		Normal	Line regularity-based			
	SP [8] +GS [29]	SP [8]+L [39] +GS [29]	U-ARE -ME [26]	H-VP [28]	1-Line RANSAC* [28]	MWMS [14]	Ours
S-Build.	2.92	25.78	8.14	2.55	1.20	0.33	0.58
N-Build.	18.08	33.96	27.05	2.30	179.55	1.20	0.62

We color the **best** and **second best** results for each sequence.

SP: SuperPoint [8], GS: GlueStick [29], L: line [39]

1-Line RANSAC* [28]: single-line solver with confident gravity prior.

Sequences. For the pure-rotation sequences (S-, N-, and C-Buildings), we report closed-loop consistency error (CLE), computed as the angular difference between estimated rotations at repeated visits to the same viewpoint. CLE measures orientation consistency without requiring per-frame rotation GT. For the 6-DoF motion sequences (S1 Building, Residential-opened, and Residential-closed), we report mean per-frame ARE against pseudo-GT rotations.

4.2 Short Pure Rotational Motion Sequences

Tab. 3 reports closed-loop consistency error (CLE) on the two shorter pure-rotation sequences, S- and N-Buildings.

Feature matching [29] and neural normal [26] estimators exhibit large closed-loop inconsistencies, reaching tens of degrees on N-Buildings. In contrast, MWMS and our method achieve substantially lower errors, with our method remaining below 1° on both sequences, while H-VP yields errors above 2° . Despite using the same gravity prior and line source as ours, 1-Line RANSAC exhibits a

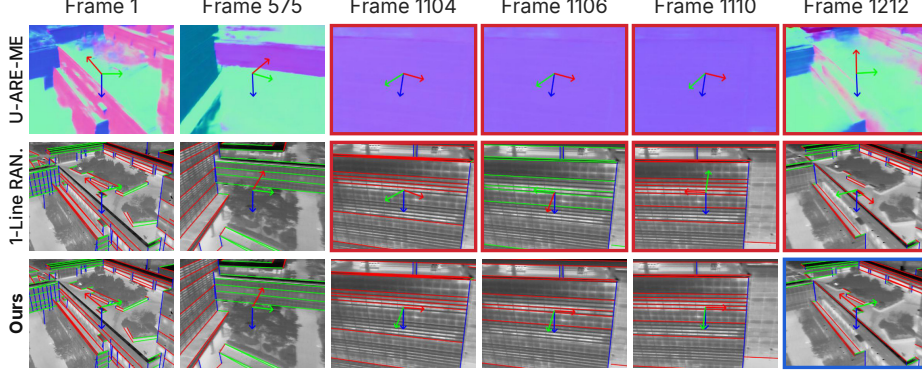


Fig. 6: Per-frame rotation estimation on N-Buildings over one full rotation, ending at the initial viewpoint. Comparing frames 1 and 1212 reveals orientation inconsistency: 1-Line RANSAC exhibits a Manhattan-axis flip, and U-ARE-ME shows misalignment, while our estimates remain consistent. Red boxes highlight misalignment; the blue box highlights consistency upon returning to the initial viewpoint.

Manhattan-axis flip on N-Buildings, whereas our MnS-based estimates remain consistent upon returning to the initial viewpoint.

The advantage of the gravity-anchored MnS strategy is reflected in both the quantitative results and the qualitative consistency shown in Fig. 6. Averaging the two sequence-level CLE values in Tab. 3 gives 0.60° for our method and 0.77° for MWMS. Fig. 6 shows a Manhattan-axis flip in 1-Line RANSAC and orientation misalignment in U-ARE-ME, while our estimates remain consistent in the illustrated frames. The MnS search aggregates support from multiple line intervals rather than relying on a single-line hypothesis.

4.3 Long Pure Rotational Motion Sequences

We evaluate long-term orientation consistency on C-Buildings, which contains 25 full rotations over 411 s (approximately 6.9 minutes). When the drone returns to its initial viewpoint, our independently estimated Manhattan frames remain consistent, with a closed-loop consistency error (CLE) of 0.287° (Fig. 7). By contrast, the DJI onboard AHRS exhibits approximately 15° of yaw drift between the two observations.

4.4 6-DoF Motion Sequences

We compare against classical feature-based [3, 30, 36], deep dense [37], geometry-aware GS-based [43], and thermal-specific GS-based [12] monocular SLAM baselines. In Tab. 4, line-regularity methods share our onboard gravity prior, whereas matching and SLAM baselines use monocular thermal input without this prior. For baselines with incomplete tracking, ARE is computed over the tracked subset.

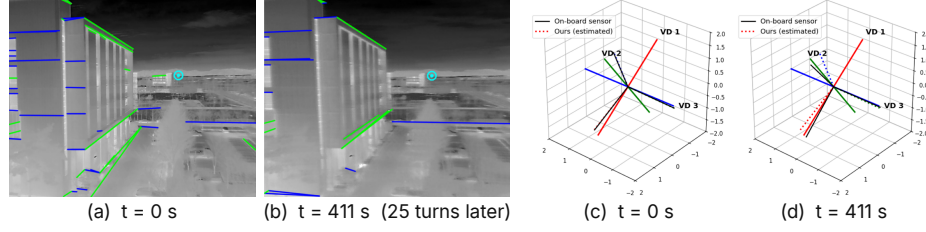


Fig. 7: Long-term orientation consistency on C-Buildings after 25 full rotations over 411 s. (a,b) Thermal images show the drone returning to its initial viewpoint. (c,d) Vanishing directions derived from the DJI onboard orientation (solid black) deviate from the initial Manhattan frame (solid red, green, and blue) due to approximately 15° of yaw drift. In contrast, our estimated directions (dotted red, green, and blue) remain nearly aligned with the initial frame, with a CLE of 0.287° .

Table 4: ARE ($^\circ$) on the three 6-DoF thermal sequences.

Sequence	Matching		Normal	Line regularity				Classical SLAM			Dense SLAM		Thermal SLAM
	SP+GS [8, 29]	SP/L+GS [8, 29, 39]	U-ARE-ME [26]	H-VP [28]	1-Line RANSAC* [28]	MWMS [14]	Ours	PL-SLAM [30]	ORB-SLAM3 [3]	Structure PLP-SLAM [36]	DROID SLAM [37]	HI-SLAM2 [43]	PI-EAT SLAM [12]
Resid.-c.	66.45	84.45	38.98	30.08	0.66	0.97	0.42	×	1.59 [†]	×	1.58 [†]	1.65	112.29
Resid.-o.	48.05	99.61	3.70	2.19	15.69	1.10	0.70	×	×	×	3.27 [†]	8.98 [†]	7.22
S1 Building	96.85	90.12	5.94	2.15	0.47	0.73	0.44	×	2.42 [†]	×	0.40	0.41	2.87 [†]

Best, second, third; ×: init. failure; [†]: before tracking failure.

Resid.-c./Resid.-o.: Residential-closed/opened; *: confident gravity prior.

Inter-frame association methods exhibit large errors, while several SLAM baselines fail to initialize or lose tracking; monocular SLAM can also degenerate under pure rotation due to insufficient parallax. DROID-SLAM and HI-SLAM2 achieve slightly lower ARE than ours on S1 Building, whereas ours achieves the lowest ARE on both Residential sequences (0.42° and 0.70°). PI-EAT-SLAM completes tracking on Residential-opened, exhibits a large rotation error on Residential-closed, and loses tracking on S1 Building. These results highlight the robustness of independent per-frame rotation estimation when inter-frame association becomes unreliable.

We additionally evaluate four SLAM baselines on Residential-closed using the same onboard gravity vector as ThermalManhattan. Keyframe roll and pitch are realigned before bundle adjustment.

Table 5: ARE ($^\circ$) on Residential-closed with and without the onboard gravity prior.

Gravity prior	ORB-SLAM3	DROID-SLAM	HI-SLAM2	PI-EAT-SLAM	Ours
Without	1.59 [†]	1.58 [†]	1.65	112.29	—
With	1.40 [†]	1.55 [†]	1.68	15.27 [†]	0.42

Most SLAM baselines improve only slightly or remain nearly unchanged with the same gravity prior, while our method still outperforms all tested SLAM baselines. This additional comparison on Residential-closed preserves the overall

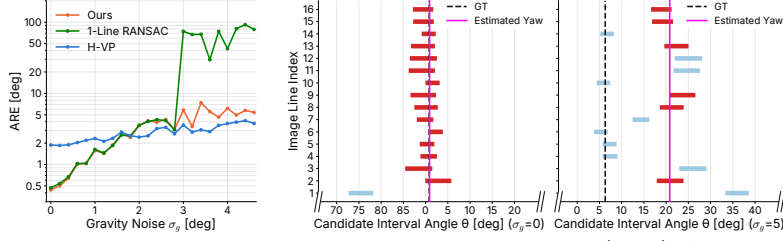


Fig. 8: Sensitivity to gravity-vector noise on S1 Building. **(Left)** ARE versus the perturbation standard deviation σ_g . **(Middle, Right)** Examples of MnS candidate yaw intervals at $\sigma_g = 0^\circ$ and 5° . Red intervals contain the selected yaw.

performance trend in Tab. 4. The gain thus comes not from the gravity prior alone but from effectively exploiting the MW structure.

4.5 Robustness Study: Gravity-Vector Noise

We evaluate sensitivity to gravity-prior noise on S1 Building. Following [28], we perturb the gravity vector by a rotation whose axis is sampled uniformly on the unit sphere and whose angle follows $\mathcal{N}(0, \sigma_g^2)$. Fig. 8 shows the resulting ARE and examples of the MnS candidate intervals. Our method achieves lower ARE than H-VP for $\sigma_g \lesssim 1.8^\circ$. For larger perturbations, H-VP is more robust, while our ARE increases without the abrupt failures observed for 1-Line RANSAC. The interval examples illustrate how gravity perturbations alter the yaw hypotheses derived from the detected lines.

4.6 Ablation Studies: Residual Geometry and Cauchy Loss

We ablate two design choices in the refinement stage, residual geometry (image plane vs. Gaussian sphere) and loss (L2 vs. Cauchy), in Tab. 6. The spherical residual consistently outperforms the image-plane residual under both losses and provides the main improvement, reducing mean ARE from 0.698° to 0.521° with L2 and from 0.682° to 0.518° with Cauchy. Cauchy down-weights high-residual inliers that can arise from endpoint jitter in thermal lines, whereas the quadratic L2 loss is more sensitive to these residuals. The additional gain from Cauchy with the spherical residual is modest, reducing mean ARE from 0.521° to 0.518° . We adopt the spherical residual with Cauchy loss, which achieves the lowest mean ARE in this ablation.

Table 6: Ablation of residual $\times$ loss (mean ARE ($^\circ$) of three 6-DoF motion sequences)

Loss	Image Plane Residual	Spherical Residual
L2	0.698	0.521
Cauchy	0.682	0.518

4.7 Application: Rescuing SLAM at Degeneracy

Absolute camera orientation is particularly useful when translation-based tracking degenerates. Residential-opened contains extended near-pure-rotation seg-

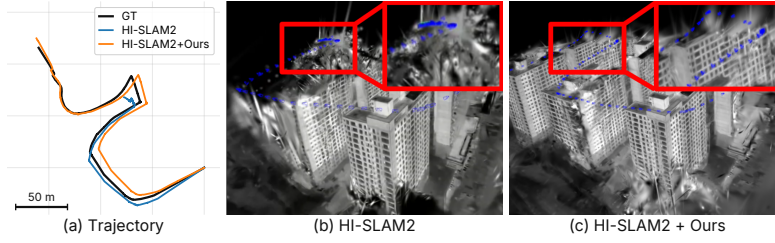


Fig. 9: HI-SLAM2 with and without ThermalManhattan on Residential-opened. **(a)** The coupled trajectory (orange) follows the reference (black) more closely than vanilla HI-SLAM2 (blue), with ARE decreasing from 8.98° over the tracked portion to 2.16° for the coupled system. **(b,c)** The coupled system better preserves building structure in the Gaussian map, as highlighted by the magnified regions.

ments without loop closure, where insufficient translational parallax causes HI-SLAM2 to drift and eventually lose tracking. Fig. 9 compares trajectories and maps with and without ThermalManhattan. At keyframes with normalized parallax below 0.02, we fix orientation to the compass estimate and rescale the tracked depth against the Gaussian map reconstructed before the degenerate segment. This coupling helps maintain tracking and preserve the reconstruction through these segments.

5 Conclusion

We present a drift-free, matching-free rotation estimation method for night-time UAV thermal imagery. Combining structural lines from a wireframe parser with gravity-anchored MnS search and spherical robust refinement yields absolute per-frame rotation without matching or tracking. In our experiments, the method stays stable where baselines fail under NUC events or pure rotation, and remains drift-free across 20+ loops while the onboard gyro accumulates approximately 15° of yaw drift. We also released the first paired RGB-thermal night-time UAV benchmark under the MW assumption, with pseudo-GT trajectories.

The principal limitations are reliance on a reliable gravity prior and sufficient Manhattan-world structure, and the lack of non-Manhattan degeneracy detection. Future work will investigate confidence estimation from MnS voting and integration with a NUC-robust thermal translation back-end for full 6-DoF estimation.

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